
AWS CloudTrail Hands-On Lab: Monitor EC2 Activity

Objective

Set up AWS CloudTrail to **track EC2 actions** (e.g., instance creation, deletion) and **analyze** them via the **CloudTrail console** or logs in **S3**.

Prerequisites

- AWS account with administrative access
 - IAM user/role with permissions to manage EC2, S3, and CloudTrail
 - Region selected (e.g., `us-east-1`)
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Step-by-Step Instructions

♦ Step 1: Create an S3 Bucket to Store CloudTrail Logs

1. Go to **S3 Console** → <https://s3.console.aws.amazon.com/s3>
2. Click **Create bucket**
3. Set:
 - Bucket name: `cloudtrail-logs-<your-unique-name>`
 - Region: Same as where you'll work

4. Leave defaults, click **Create bucket**

♦ Step 2: Create a CloudTrail Trail

1. Go to **CloudTrail Console** → <https://console.aws.amazon.com/cloudtrail>
2. Click **Trails** → **Create trail**
3. Trail name: **EC2-Traill**
4. Apply trail to **All regions** 
5. Choose **Create a new S3 bucket** (or use one you created earlier)
6. Enable:
 - **Management events** → Read & Write
 - **Data events** → Skip for now (not needed for EC2)
7. Click **Next** → **Create trail**

CloudTrail will now start recording all EC2-related actions (and others).

♦ Step 3: Perform EC2 Activities

Go to **EC2 Console** → <https://console.aws.amazon.com/ec2>

1. **Launch an EC2 instance**
 - Use Amazon Linux 2 AMI
2. **Stop and Start** the instance
3. **Terminate** the instance

These actions generate **CloudTrail events**.

♦ Step 4: View CloudTrail Events (Console)

1. Go to **CloudTrail Console** → **Event history**
2. Filter by:
 - **Event source:** `ec2.amazonaws.com`
 - OR **Event name:** `RunInstances`, `StopInstances`, `TerminateInstances`
3. You'll see:
 - Time of action
 - IAM user/role who performed it
 - Resource ID (e.g., instance ID)
 - Source IP

Click on an event for full JSON details.

♦ Step 5: View Raw Logs in S3

1. Go to the **S3 bucket** you created

Navigate folders:

`AWSLogs/<AccountID>/CloudTrail/<region>/<year>/<month>/<day>/...`

- 2.
3. Download and open a `.json.gz` log file

Use a viewer or `jq` to parse:

```
zcat <file>.json.gz | jq .
```

4.

Bonus: Setup Alerts (Optional)

1. Send CloudTrail logs to **CloudWatch Logs**
 2. Create a **metric filter** for EC2 `TerminateInstances`
 3. Create **CloudWatch Alarm** and trigger **SNS email alert**
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Useful CloudTrail Event Names for EC2

Action	Event Name
Launch Instance	<code>RunInstances</code>
Stop Instance	<code>StopInstances</code>
Start Instance	<code>StartInstances</code>
Terminate Instance	<code>TerminateInstances</code>
Create Key Pair	<code>CreateKeyPair</code>
Modify SG	<code>AuthorizeSecurityGroupIngress</code>

Summary

Component	Purpose
CloudTrail	Logs API calls
S3	Stores logs
EC2 actions	Generate events

Event history Analyze who did what & when
